Owen Patel

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Education

Westhaven College (Fictional) *Sep. 2021 - Jun. 2025*
BSc in Applied Mathematics; Minor in Computing Manchester, UK
GPA: 3.78/4.00

Relevant Coursework: Numerical Analysis, Optimization, Differential Equations, Data Structures

Research Experience

Research Assistant - Numerical Methods *Jan. 2025 - Sep. 2026*
Westhaven Computational Science Group | Advisor: Dr. Priya Morgan

- Implemented finite-difference solvers for two-dimensional heat transport and checked convergence on analytic solutions.
- Compared conjugate-gradient and multigrid solvers over four grid resolutions with fixed stopping criteria.
- Measured a 2.1x runtime reduction for multigrid on the largest grid; reported setup cost and memory tradeoffs.
- Used manufactured solutions to validate boundary handling and estimate empirical convergence rates.
- Profiled sparse matrix assembly separately from iterative solution to identify where runtime improvements originated.
- Analyzed stopping tolerances, conditioning and residual error rather than treating speed as the only objective.
- Wrote a 24-page internal report describing discretization assumptions, numerical stability and reproducibility.

Undergraduate Thesis - Optimization under Noise *Sep. 2024 - Jun. 2025*
Westhaven Applied Mathematics Department | Advisor: Dr. Samuel Park

- Compared stochastic gradient and variance-reduction methods on synthetic convex objectives.
- Reported sensitivity to step size and noise over 30 independent runs; produced a departmental thesis.
- Derived baseline convergence expectations for smooth objectives and compared them with empirical trajectories.
- Performed controlled experiments on batch size and noise variance, recording function evaluations as well as iterations.
- Separated failed runs from successful runs and included sensitivity plots in the thesis appendix.

Professional Experience

Westhaven College *Sep. 2024 - May 2025*
Numerical Methods Teaching Assistant Manchester, UK

- Led weekly Python problem sessions and reviewed solver correctness for 40 students.
- Prepared exercises on floating-point error, iterative solvers and numerical differentiation.
- Used reference solutions and unit tests to help students diagnose indexing and boundary-condition errors.
- Maintained an anonymized issue log and revised examples after recurring misconceptions.

Selected Projects

Sparse Linear Algebra Benchmark Suite *2025 - 2026*
Python, SciPy, C++, CMake

- Built benchmark scripts for runtime, residual error and memory use across sparse matrix families.
- Published reproducible instructions and checks against reference solutions in a demonstration repository.
- Included warm-up runs, repeated timing and explicit matrix sizes to reduce misleading benchmark comparisons.
- Recorded solver failures and nonconvergence alongside successful timings.

- Added a command-line interface and documented build flags, thread counts and reference tolerances.

Optimization Learning Notebook Series *Jun. 2025 - Dec. 2025*

Python, Jupyter, NumPy, Matplotlib

- Created six teaching notebooks covering gradient methods, conditioning and constrained optimization.
- Used synthetic counterexamples to demonstrate unstable step sizes and misleading stopping criteria.
- Added reproducible figures and exercises comparing analytic and numerical gradients.
- Collected feedback from 12 classmates and revised explanations of convergence diagnostics.

PDE Solver Verification Lab *2024*

MATLAB, Python, NumPy

- Implemented explicit and implicit time-stepping methods for one-dimensional diffusion.
- Compared stability and error under controlled changes in grid spacing and timestep.
- Verified boundary conditions using manufactured solutions and conservation checks.
- Created plots distinguishing discretization error from floating-point effects.
- Prepared reproducible notebooks explaining assumptions to classmates without a numerical-analysis background.

Algorithmic Thinking Study Group *2025*

C++, Python, Git

- Organized an eight-week peer study group connecting mathematical proofs with executable algorithms.
- Implemented graph traversal, dynamic programming and sparse matrix examples with reference tests.
- Reviewed complexity arguments and documented differences between worst-case bounds and observed runtimes.
- Maintained a shared repository with reproducible exercises and contributor notes.
- Collected anonymous feedback and revised exercises that confused mathematical notation with implementation details.

Technical Skills

Programming Languages: Python, C++, MATLAB

Frameworks & Libraries: NumPy, SciPy, Matplotlib

Systems & Tools: Linux, Git, CMake

Specializations: Numerical linear algebra, optimization, scientific computing

Honors & Awards

Westhaven Mathematics Dissertation Prize 2025 – *Recognized a reproducible experimental thesis.*

Computational Modeling Team Challenge 2026 – *Second place in a fictional college team modeling competition.*